

Exercise 9

AIM :

To create an array using dynamic memory allocation and sort the array using user defined function.

ALGORITHM :

1. START
2. INCLUDE `stdio.h` and `stdlib.h`
3. DECLARE FUNCTIONS:
 - `inputArray(int *arr, int size)`
 - `displayArray(int *arr, int size)`
 - `sortArray(int *arr, int size)`
4. MAIN FUNCTION:
 - a. DECLARE: `size (int), *arr (int pointer)`
 - b. INPUT: Get array size from user
 - c. ALLOCATE: `arr = malloc(size * sizeof(int))`
 - d. IF allocation fails: print error and `EXIT(1)`
 - e. CALL `inputArray(arr, size)`
 - f. PRINT "Original " + CALL `displayArray(arr, size)`
 - g. CALL `sortArray(arr, size)`
 - h. PRINT "Sorted " + CALL `displayArray(arr, size)`
 - i. FREE memory: `free(arr)`
 - j. RETURN 0
5. `inputArray` FUNCTION:
 - a. PRINT "Enter size integers:"
 - b. FOR `i=0` to `size-1`:
 - i. PRINT "Element `i+1`: "
 - ii. READ `arr[i]`
 - c. END FOR

6. displayArray FUNCTION:

- a. PRINT "Array elements are:"
- b. FOR i=0 to size-1:
 - i. PRINT arr[i] + space
- c. END FOR
- d. PRINT newline

7. sortArray FUNCTION (Bubble Sort):

- a. FOR i=0 to size-2:
 - i. FOR j=0 to size-i-2:
 1. IF arr[j] > arr[j+1]:
 - a. temp = arr[j]
 - b. arr[j] = arr[j+1]
 - c. arr[j+1] = temp
 - ii. END FOR
- b. END FOR

8. END

OUTPUT :

```
PS D:\C programming\DHAVAMANI_EX9> gcc main.c -o main
PS D:\C programming\DHAVAMANI_EX9> ./main.exe
Enter the number of elements in the array: 5
Enter 5 integers:
Element 1: 98
Element 2: 87
Element 3: 76
Element 4: 54
Element 5: 43
Original Array elements are:
98 87 76 54 43
Sorted Array elements are:
43 54 76 87 98
```